

The University of Notre Dame

Response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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Introduction

The University of Notre Dame is pleased to provide a comprehensive response to the Office of Science and Technology Policy's (OSTP) request for input on the development of a national artificial intelligence (AI) action plan. In our response, we will address key priorities for advancing AI research and policy in alignment with the OSTP's objectives, focusing on five core categories: research, workforce development, safety, ethics, and policy. We recognize the growing concerns across various sectors about the potential risks associated with AI, including bias, inequitable access, safety vulnerabilities, and ethical challenges. In light of these concerns, we propose suggestions for an action plan that will guide the responsible, inclusive, safe, and ethical development of AI technologies, while ensuring that AI remains a tool for enhancing human well-being, empowering individuals, and advancing societal progress. Through our detailed recommendations in the areas of research, workforce development, safety, and policy, we aim to contribute to shaping a future where AI is developed and deployed in a manner that benefits all.

Research

We have identified various opportunities for research advancement in the field of AI. It is critical that the United States maintains its leadership in advancing both fundamental and applied AI research. This requires a concerted effort from government, industry, and academia to foster development of new algorithms, new computing paradigms, accelerate scientific discovery, and identify innovation pathways to spur new start-ups. Continuing to bolster the American research ecosystem will ensure that the United States remains at the forefront of innovation.

1. (Opportunity) Open-source all federally funded fundamental research content. We advise the creation of a framework for AI governance to allow each federally funded program to clearly state the parameters of open source disclosure of fundamental research content (papers, software, data). The program will provide a risk management matrix for such research. Making this content AI-accessible would accelerate discovery across all fields.
2. (Opportunity) Small (<\$1M) and midscale research infrastructure (<\$20M) grants for AI-enablement. We are in the process of discovering what an

AI-enabled laboratory looks like - it's clear that it varies by domain, involves more data capture, and involves more automation. Researchers and institutions need the means to experiment and reimagine research spaces.

3. (Opportunity) Investments in reproducible research. AI may dramatically improve scientific reproducibility by automating experimental procedures, standardizing data collection, and performing verification and contradiction resolution which are poorly incentivized in the current research culture. Pilots for building these systems and integrating them with different modes of research need to be launched.
4. (Opportunity) Investments in fundamental research to create new foundational AI models that can work with lower compute and lower data footprint. It is essential that academia focuses on approaches such as knowledge distillation, federated learning, domain-informed fine tuned language models, explainability, and transparency. It will not be possible for academia to compete with private sector investments in computing infrastructure and data.
5. (Opportunity) Investments in forging a public-private partnership that allows dedicated compute and data sandboxes for co-innovation. The United States research ecosystem has excelled due to the academia-industry-government co-innovation ecosystems. It is critical to double down on such existing frameworks to accelerate research progress and create pathways for fundamental research to innovation to translation in AI.
6. (Opportunity) Edge and Federated AI advances to develop AI methodologies to work on edge devices and also in a federated framework at remote sites such that data can be protected. Applicable to various strategic areas such as drones, robotics, scientific discovery, next-generation wireless technologies (e.g., 6G and beyond), etc.
7. (Opportunity) Promote the development and deployment of AI-powered robotic systems in critical sectors including healthcare, infrastructure, manufacturing, agriculture, and disaster response to enhance safety, productivity, and adaptability.
8. (Opportunity) Establish Responsible Safe and Ethical AI guidelines for informing the current and next generation of AI systems. This includes establishing testbeds and regulatory sandboxes to safely evaluate and accelerate the deployment of robotic technologies in real-world environments.

Beyond identifying areas for growth, we have also determined four key risk areas that threaten innovation and technological advancement.

1. (Risk) Obsolete culture and methods for domain assurance and validation. As we move away from interpretable models towards black-box predictions, establishing trust, validating results, and maintaining scientific rigor becomes increasingly difficult. New frameworks for verification and validation must be developed.
2. (Risk) Ad hoc institutional responses. Current approaches to AI integration in research lack coherent, systematic frameworks. Institutions need to develop sustainable strategies that address domain assurance, workforce development, and infrastructure challenges simultaneously.
3. (Risk) Incomplete guidance and framework for open source of AI software and data creates a lack of a consistent reproducibility and repeatability rigor. It will be important to also establish standard benchmarks.
4. (Risk) Insufficient access to computational and data resources can hamper research in both basic and applied AI.

Workforce Development

New AI technologies are expected to have a significant impact on future manufacturing operations, workers, and jobs. While most people believe AI technologies will naturally displace these jobs, we believe AI technologies should be viewed as another tool available to the worker. Similar to the way computer numerical controls (CNCs) replaced hand lathes to not only improve efficiency but relieve workers from physically demanding and potentially dangerous work, AI technologies have the potential to improve efficiency while relieving workers of repetitive tasks and providing them with highly desirable jobs in which they can unleash their creativity by working collaboratively with these technologies.

Industry is rapidly evolving with the adoption of AI and emerging technologies. In order to ensure a pipeline of workers to maintain America's AI dominance, we must create workforce development programs that merge hands-on learning with theoretical knowledge and practical skills, including proficiency in AI tools. We must seek to address the widening skills gap by cultivating a workforce capable of interdisciplinary, creative problem-solving, ensuring that both emerging professionals and current practitioners are well-equipped to drive societal transformation in an era of rapid technological change.

Where gaps in the labor force emerge as a result of the integration of AI, we must ensure that there is wholehearted commitment to programs that promote upskilling, or opportunities to increase and advance skills in a targeted area. In focusing on ameliorating skill gaps among existing workers, the United States can remain dominant in the field of AI, while ensuring that the nation's workforce is strong and well-equipped to adopt technological advances.

Furthermore, when considering a longitudinal view of the nation's workforce abilities, it becomes clear that, in order to responsibly cultivate the next generation of workers, it is our responsibility to prepare students for a future where AI becomes increasingly powerful and plays important roles in productivity tasks and in our daily lives, not least by cultivating in them the skills necessary for effective and ethical human-AI collaboration. We advocate for a curriculum that not only incorporates AI tools, but also critically examines their limitations and potential affordances. This approach has the potential to cultivate a new generation of workers who are not only precise and proficient in their use of generative AI, but also conscientious about the ethical dimensions of these technologies.

Despite much discussion and debate surrounding AI technologies, jobs and tasks, workers and skill, and labor market constraints and inequality in manufacturing, there has been no in-depth study that examines the future of work of factory floor workers holistically from an integrated economic, technological, organizational, and sociological framework. Developing a framework where manufacturing firms, including small and medium-sized enterprises, can use when adopting AI technologies so that they may symbiotically grow as workers and firms in their local economies will be important.

Below we outline opportunities and risks that could be explored in such a study.

1. (Opportunity) **Workforce transformation enablement.** Large swathes of professional work are already obsolete, creating a massive opportunity for redeploying and equipping the current workforce to redesign their industries. We can expect a transformation comparable to the entire digital revolution but concentrated into a decade.
2. (Opportunity) **Enhanced skill acquisition and expert knowledge dissemination.** Training is better than ever with AI providing immediate feedback and personalized instruction that accelerates learning. Complex skills can now be mastered in months rather than years. Simultaneously, AI enables the systematic capture and dissemination of tacit knowledge from experts in universities and institutions before it's lost to retirement, creating unprecedented knowledge transfer across generations.
3. (Opportunity). **Interdisciplinary and experiential learning opportunities for both undergraduate and graduate students.** It is important to ensure that the students are equipped not only to have a deep understanding of fundamental AI (spanning algorithms to underlying computing architecture) but also have an experience in applying AI to solve domain specific problems.
4. Risk) **Programs to offset occupational obsolescence.** Comprehensive upskilling of the current generation of workers may be necessary over the next

decade to keep pace with AI integration. Those unable to adapt may find themselves increasingly marginalized leading to large social dislocations.

5. (Risk) **Traditional training models are too slow and costly.** AI advances are on pace to require massive upskilling and redeployment of the current workforce, but our traditional training infrastructure is inadequate for the scale and speed of transformation needed. Educational institutions and corporate training programs must evolve rapidly.

Responsible, Inclusive, Safe, and Ethical AI

Given the rapidly emerging nature of AI, stakeholders across all sectors have expressed significant concerns about bias, inequitable access, safety vulnerabilities, and ethical uncertainties of AI. Not to be interpreted as opposition to the integration of AI, rather, the concern for responsible deployment of a technology as expansive and with as much potential as AI demonstrates an understanding of its power and a sense of responsibility to ensure it is deployed thoughtfully. This highlights the urgent need for a guiding framework for the role of AI at the interface of different domains and disciplines while ensuring that AI technologies are developed and applied responsibly, inclusively, safely, and ethically.

When considering necessary guardrails for the promotion of AI safety, we believe that there are four aspects that should be considered—AI as Responsible, Inclusive, Safe, and Ethical (RISE).

We see the following research and training areas as key priorities for sustaining AI dominance in the United States while prioritizing safety and responsibility.

- **Responsible:**
 - Designing, developing, deploying, and studying interactive AI systems to enhance human-AI collaboration, augment human intelligence, and democratize access to AI empowerment technologies in a responsible way.
 - Mechanisms for measuring, evaluating, and informing users of the potential privacy and safety risks associated with the use of Large Language Models (LLMs) and LLM-powered agents, especially in critical domains e.g., health care, social services.
 - AI Tools that improve AI literacy of end users and help them make informed decisions aligned with their own values and long-term interests.
 - Policies to standardize data sharing formats and practices for pharmaceutical companies and organic chemists more broadly.
- **Inclusive**

- Designing, developing, and deploying AI systems that bridge instead of widen the existing economic disparities in the US.
- Safe:
 - Interdisciplinary research to advance the trust and reliability of AI when applied to systems involving humans (such as biometric recognition) and serving humans (such as AI-assisted systems aiding human experts).
 - Testbeds and sandboxes for assessing and benchmarking the safety and trustworthiness of AI models/agents.
- Ethical
 - Systems for the measurement, evaluation, and alignment of model and agent behaviors to human values.
 - Creating interpretable, advanced AI-driven models must foster rational, proactive, and ideology-free policy-making across multiple governance domains, including public policy, corporate governance, and regulatory frameworks. By embedding ethical safeguards, rigorous validation, and continuous oversight, these models can enhance decision-making processes, mitigate systemic risks, and ensure equitable and accountable outcomes in an increasingly data-driven world.

Furthermore, the University of Notre Dame is dedicated to ensuring that AI deployment aligns with global human rights norms and democratic principles.

An Ethics-Based Framework for AI

We believe that Notre Dame has a unique role to play in convening, leading, and amplifying the voices of leaders in higher education, technology, and faith-based communities who are invested in developing faith-based ethical frameworks and applying them to emerging debates around AI. At this time, the University is creating a powerful network of innovative, influential leaders who will be actively immersed in this work.

We strongly encourage the government to be deliberate and thoughtful in its inclusion of voices of faith in AI policy discussions and decisions so that our collective uses of this powerful new technology can move beyond compliance to the realm of human flourishing.

Through a variety of strategic research initiatives, we have come to the following initial conclusions:

- There is a great need for context-specific ethical frameworks and context-aware LLMs. The way that geopolitical conflict, for example, is portrayed by these models can have an outsized impact on moments of heightened social unrest and civil conflict in the present. In addition, local communities can benefit greatly from these powerful new technologies if their contexts and use-case needs are kept in mind by policymakers and tech companies.
- AI serves to accelerate any pre-existing dynamics related to society's use of digital technologies. Therefore, there is a greater need than ever to double down on commitments to transparency, the right to privacy, and individuals' fundamental ownership over their own data.
- Existing frameworks and standards for AI governance are not always easily transferable to new contexts or rapid developments in the technology itself. This means that individual businesses, sectors, and communities will need to be equipped to establish apt governance of emerging AI technologies at the appropriate level (whether international, federal, state, or local) and in the relevant industry or domain. Material investments in technology infrastructure and access will need to be made to achieve this goal.

Policy

Effective AI development depends heavily on securing superior data, robust computing infrastructure, and highly skilled talent. Therefore, policies should actively nurture AI expertise within the U.S. education system while also incentivizing talented individuals educated in the U.S. to remain and innovate domestically.

Based on our research and expertise, as well as our collaborations with global AI governance experts, we propose the following key policy actions:

1. Establish a National AI Trust Framework that incorporates blockchain, decentralized identifiers (DIDs), and verifiable credentials (VCs) to ensure secure AI model and data provenance tracking while preserving privacy.
2. Enhance AI cybersecurity protections by integrating privacy-preserving mechanisms such as zero-knowledge proofs (ZKPs), federated learning, and robust AI security standards to mitigate adversarial threats and safeguard AI systems from data poisoning and manipulation.
3. Develop AI governance models that leverage decentralized, transparent systems, reducing the risks associated with centralized control over AI technologies. Our research in blockchain governance, smart contract accountability, and decentralized AI governance through the NSF Cyber SMART Project and BGIN supports this initiative.

4. Align AI policy with global human rights frameworks, ensuring that AI adoption in government, industry, and academia adheres to international ethical standards.
5. Regulate AI's role in corporate governance and accountability, recognizing that large technology corporations wield significant influence over AI policy and deployment.
6. Invest in AI education and workforce development, ensuring a diverse talent pipeline capable of responsible AI innovation. This includes expanding interdisciplinary AI governance research, supporting AI ethics training, and integrating AI literacy into STEM and policy education.
7. Significantly expand AI computing resources for U.S. science and research. The United States must make substantial, strategic investments in AI computing infrastructure to support scientific discovery, innovation, and national competitiveness. The National Artificial Intelligence Research Resource (NAIRR) is a commendable initiative toward democratizing access to AI resources; however, it requires a more aggressive budget to meet the growing demands of the research community. The NAIRR Task Force's final report, published in January 2023, estimated that \$2.6 billion would be required to operate the NAIRR over six years, with \$2.25 billion allocated for resource availability across multiple federal agencies. Despite this, the current budget request includes only \$30 million for the second year of the NAIRR pilot, a figure that falls significantly short of the necessary investment. Without adequate funding, the NAIRR cannot effectively bridge the gap between resource-rich corporations and academic or smaller research entities, potentially hindering innovation and equitable access to AI advancements. We strongly urge the government to prioritize and substantially increase funding for the NAIRR and related AI infrastructure projects to ensure that AI advancements are not monopolized by a select few but are accessible to a broad spectrum of researchers and institutions.
8. Invest in sustainable and energy-efficient hardware design for scaled AI, allowing the United States to facilitate larger scale deployment and training of future AI systems, enabled by research and investments in low-carbon, energy-efficient AI hardware development. Specifically, we recommend investment in Chips-for-AI programs to create specialized hardware and techniques optimized for AI, significantly reducing energy consumption and carbon footprints. We recommend pairing directed investment with the use of program incentives to streamline semiconductor design and manufacturing processes, enhancing innovation, reducing waste, and furthering sustainability goals. We believe that

encouraging robust partnerships between industry and academia will accelerate the commercialization of sustainable hardware innovations.

These aforementioned policy actions leverage existing research expertise and align with national objectives on sustainability, technological advancement, responsible AI governance, and equitable workforce development.

Our research and global collaborations directly support the U.S. government's goal of maintaining AI leadership while ensuring security, privacy, human rights, and ethical responsibility. We welcome continued engagement in shaping AI policy at the national and international levels.

Collaborators at the University of Notre Dame

Individuals that contributed to this Request for Information include representatives from [the Keough School of Global Affairs](#), [the Center for Research Computing](#), [the Lucy Family Institute for Data and Society](#), [the Institute for Ethics and the Common Good](#), and [the College of Engineering](#).